

AMENDMENT NO. 1 JULY 2022
TO
IS 4998 : 2015 DESIGN OF REINFORCED CONCRETE CHIMNEY —
CRITERIA
(*Third Revision*)

(Page 2, clause 3, symbol H) — Insert the following entry after H :

‘ H ’ = Total height of the chimney above top of the foundation (m) ’.

(Page 3, clause 3, symbol β_2) — Insert the following entry after β_2 :

‘ β_{sum} = sum of the angles of all openings (deg)’.

(Page 4, clause 5.3) — Substitute the following for the existing:

‘Earthquake loads on chimneys shall be computed in accordance with IS 1893 (Part 4)’.

(Page 4, clause 5.5.3, line 11) — Substitute the following for the existing:

‘ k_4 shall be taken as 1.15 for chimneys located in cyclonic regions as recommended in IS 875 (Part 3) and shall be taken as 1.00 for chimneys located in other regions.’

(Page 7, clause 5.5.8, heading) — Substitute ‘*Natural Frequencies for the Evaluation of Wind Loads*’ for ‘*Natural Frequencies*’.

[Page 8, clause 6.2.1 (c), last line] — Substitute ‘0.07’ for ‘0.05’.

(Page 12, clause 6.3.1) — Substitute the following for the existing:

‘The maximum lateral deflection at the top of the chimney due to wind loads under all service conditions with $\gamma_f = 1.0$ shall be limited to $H/500$, where H is the total height of the chimney above the top of the foundation. In the deflection calculation, modulus of elasticity of concrete (E_c) as per IS 456 shall be used with the assumption of uncracked section and fixed base. With regard to earthquake forces, reference should be made to IS 1893 (Part 4).’

[Page 13, clause 6.4.3.4 (b)] — Insert ‘(not cellular)’ after ‘solid’.

(Page 13, clause 7, first para) — Substitute the following for the existing:

‘The one-half opening angle of any opening shall not exceed 30° . The sum of the angles of all openings (β_{sum}) at any cross-section shall not exceed 72° .’

(Page 16, clause B-1) — Substitute the following formula for P_{umax} for the existing:

$$P_{\text{umax}} = (2\pi r t) \left(1 - \frac{\beta_{\text{sum}}}{360} \right) \left[\frac{0.67 f_{\text{ck}}}{\gamma_c} \left(1 - \frac{p}{100} \right) \right] + (2\pi r t) \left[f_s(\epsilon_{\text{cu}}) \frac{p}{100} \right]$$

where

β_{sum} = Sum of the angles of all openings (deg)

(Page 16, Table 2, last row, column 2) — Substitute ‘Up to 0.07’ for ‘Up to 0.05’.

(CED 38)

Publication Unit, BIS, New Delhi, India